

Four-Database Graph Benchmark

Performance Benchmarking of CognoDB, Neo4j Aura, Memgraph, and FalkorDB Using Graph Workloads

Project	Four-Database Graph Benchmark
Databases	CognoDB, Neo4j Aura, Memgraph, FalkorDB
Dataset	Pokec 100K-edge processed graph dataset
Latency Runs	100 runs per workload/database
Mixed Workload	80% Read / 20% Write
Concurrency	1, 10, and 40 workers
Date	21 August 2026

This report summarizes the completed experimental benchmark comparing four graph database systems across aggregation, indexed lookup, graph traversal, and mixed read/write workloads.

1. Introduction

Graph databases are designed to represent entities and relationships and to efficiently support operations such as graph traversal, indexed lookup, aggregation, and concurrent workloads. This project evaluates four graph database systems under a common benchmark methodology.

The systems evaluated are CognoDB, Neo4j Aura, Memgraph, and FalkorDB. The benchmark focuses on measured latency and throughput under the selected experimental configuration rather than making a universal claim about database performance.

2. Objectives

- Compare query latency across four graph database systems.
- Measure node-count aggregation performance.
- Measure indexed point-lookup performance.
- Measure 1-hop, 2-hop, and 3-hop graph traversal latency.
- Evaluate mixed read/write throughput under different concurrency levels.
- Calculate performance gaps and speedups relative to CognoDB.
- Produce reproducible CSV summaries and performance visualizations.

3. Dataset and Experimental Setup

Parameter	Value
Dataset	Pokec 100K-edge processed dataset
Relationships	100,000
Unique nodes	169,870
Databases	CognoDB, Neo4j Aura, Memgraph, FalkorDB
Latency workloads	Aggregation, indexed lookup, 1/2/3-hop traversal
Latency repetitions	100 runs per workload
Mixed workload duration	30 seconds
Read ratio	80%
Write ratio	20%
Concurrency levels	1, 10, 40

4. Benchmark Workloads

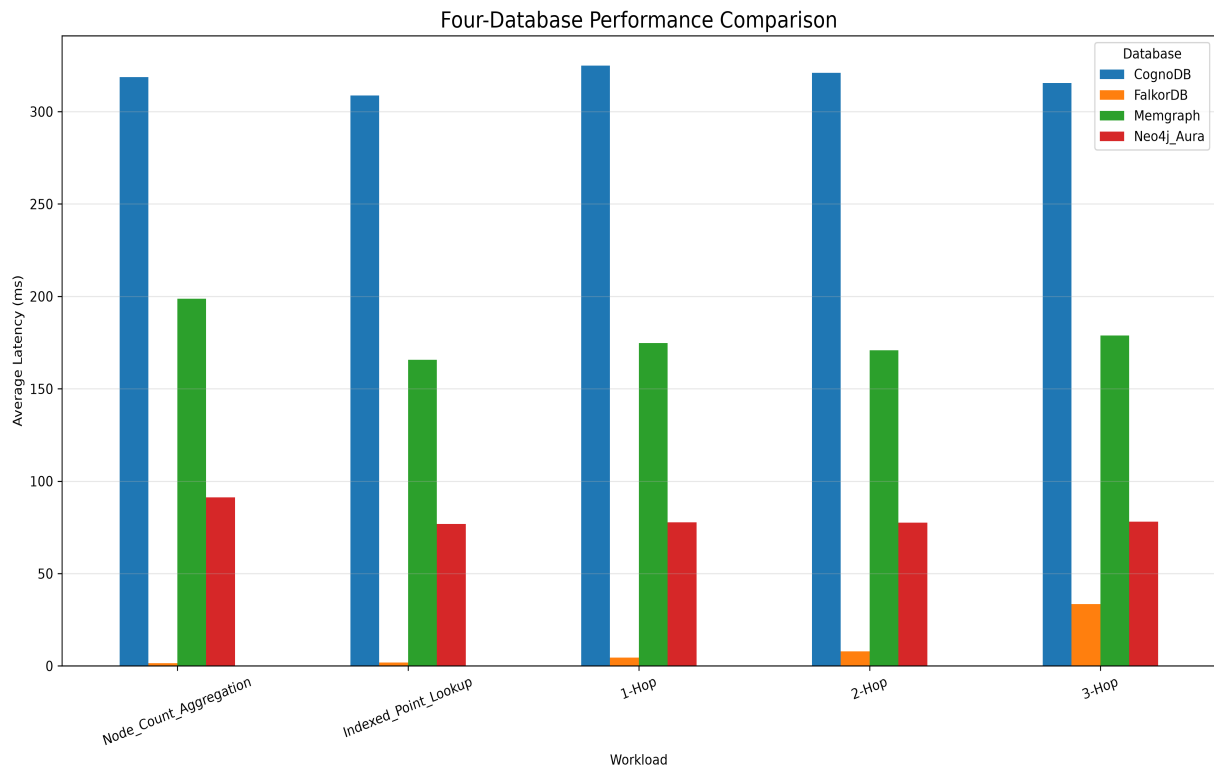
Workload	Description
Node-count aggregation	Measures the time required to count nodes in the graph.
Indexed point lookup	Retrieves a node using the indexed Node.id property.
1-hop traversal	Measures traversal from a starting node to directly connected nodes.
2-hop traversal	Measures traversal across two relationship hops.
3-hop traversal	Measures traversal across three relationship hops.
Mixed workload	Combines read and write operations using 80% reads and 20% writes.

5. Latency Results

Workload	Fastest	Avg ms	Slowest	Avg ms	Gap
Node count	FalkorDB	1.4117	CognoDB	318.7062	225.76×
Indexed lookup	FalkorDB	1.8424	CognoDB	308.7935	167.60×
1-hop	FalkorDB	4.5598	CognoDB	324.8248	71.24×
2-hop	FalkorDB	7.8119	CognoDB	320.9539	41.09×
3-hop	FalkorDB	33.5168	CognoDB	315.5224	9.41×

FalkorDB recorded the lowest average latency in all five measured latency workloads. CognoDB recorded the highest average latency for each workload. The largest measured performance gap occurred for node-count aggregation at 225.76×, while the smallest occurred for 3-hop traversal at 9.41×.

Figure 1. Four-database latency comparison



6. Detailed Latency Observations

- FalkorDB achieved 1.4117 ms average latency for node-count aggregation.
- FalkorDB achieved 1.8424 ms average latency for indexed point lookup.
- FalkorDB recorded 4.5598 ms, 7.8119 ms, and 33.5168 ms for 1-hop, 2-hop, and 3-hop traversal respectively.
- Neo4j Aura consistently showed lower latency than Memgraph and CognoDB in the reported latency workloads.
- The FalkorDB advantage decreased with increasing traversal depth, indicating that deeper traversal was comparatively more expensive.

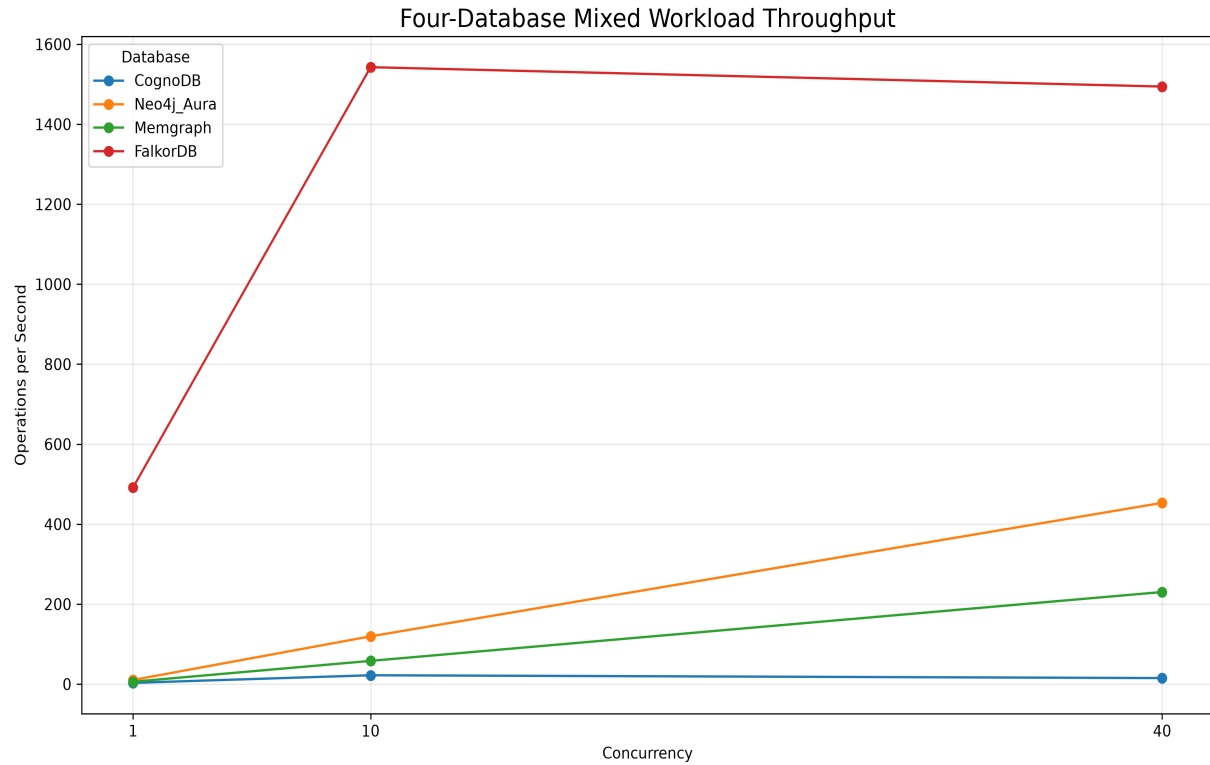
7. Mixed Read/Write Workload

Concurrency	FalkorDB	Neo4j Aura	Memgraph	CognoDB
1	491.63	10.31	6.00	3.11
10	1542.55	119.48	58.08	22.13

40	1494.16	453.23	230.27	15.07
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FalkorDB produced the highest measured throughput at every tested concurrency level. Its throughput increased from 491.63 operations per second at concurrency 1 to 1,542.55 operations per second at concurrency 10. At concurrency 40, throughput was 1,494.16 operations per second.

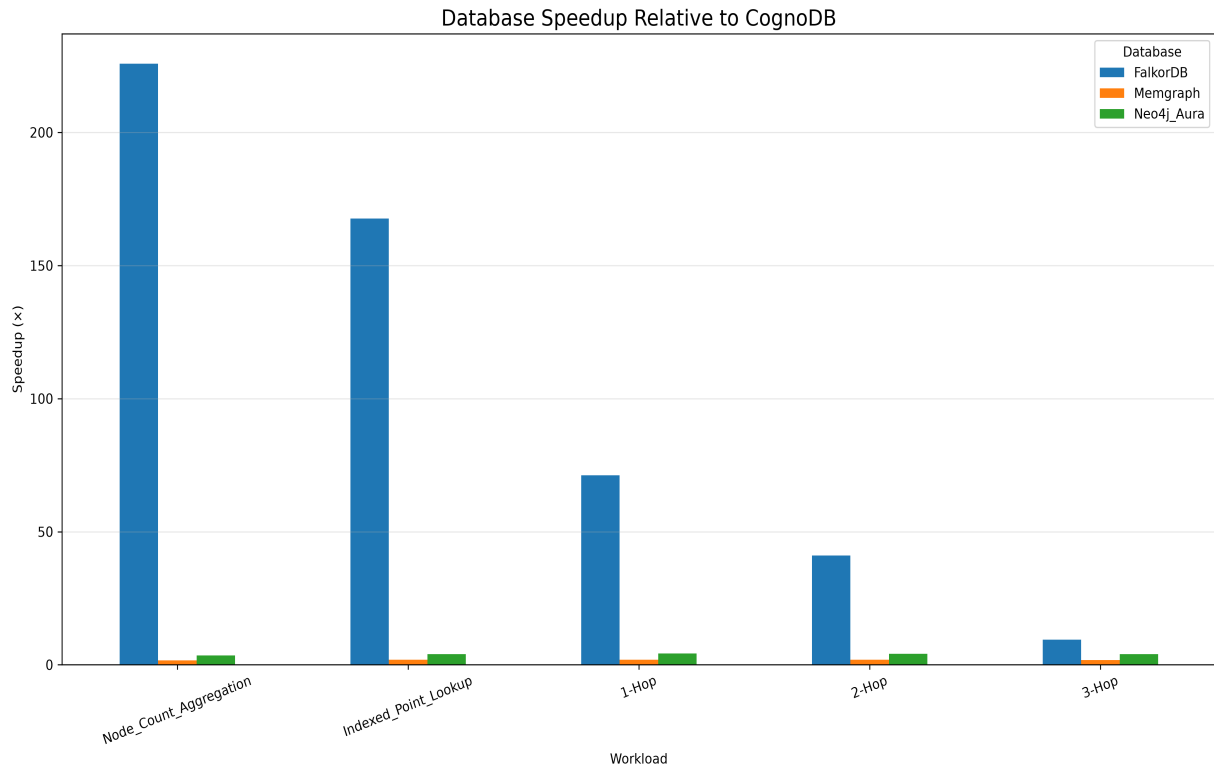
Figure 2. Mixed workload throughput comparison



8. Speedup Relative to CognoDB

Workload	Neo4j Aura	Memgraph	FalkorDB
Node count	3.49×	1.60×	225.76×
Indexed lookup	4.02×	1.86×	167.60×
1-hop	4.18×	1.86×	71.24×
2-hop	4.14×	1.88×	41.09×
3-hop	4.04×	1.76×	9.41×

Figure 3. Speedup relative to CognoDB



9. Discussion

- FalkorDB was the fastest system across every latency workload measured.
- Neo4j Aura consistently ranked ahead of Memgraph and CognoDB in the reported latency measurements.
- FalkorDB showed its largest relative advantage in aggregation and indexed lookup.
- The relative advantage of FalkorDB decreased as traversal depth increased.
- FalkorDB achieved the highest throughput under all tested mixed-workload concurrency levels.
- Neo4j Aura scaled strongly from 10 to 40 concurrent workers in the mixed workload.
- Memgraph produced measurable throughput under concurrency 40 but remained below Neo4j Aura and FalkorDB.
- CognoDB recorded the lowest measured throughput in the mixed workload.

10. Final Benchmark Summary

Workload	Fastest	Fastest Avg	Slowest	Slowest Avg	Gap
Node aggregation	FalkorDB	1.4117 ms	CognoDB	318.7062 ms	225.76×

Indexed lookup	FalkorDB	1.8424 ms	CognoDB	308.7935 ms	167.60×
1-hop	FalkorDB	4.5598 ms	CognoDB	324.8248 ms	71.24×
2-hop	FalkorDB	7.8119 ms	CognoDB	320.9539 ms	41.09×
3-hop	FalkorDB	33.5168 ms	CognoDB	315.5224 ms	9.41×

Overall latency result: FalkorDB was fastest in all five latency workloads.

Overall throughput result: FalkorDB achieved the highest measured throughput at concurrency 1, 10, and 40.

11. Limitations and Reproducibility

The benchmark results are specific to the selected dataset, workload definitions, query implementations, deployment architecture, hardware, network conditions, database configuration, indexing strategy, and client libraries. Therefore, the measured results should not be interpreted as universal rankings of the database systems.

For stronger scientific validation, future experiments should document hardware and software versions, database configurations, query plans, warm-up and cache conditions, and resource allocation. Multiple independent benchmark repetitions should also be reported.

12. Conclusion

The completed benchmark demonstrates substantial performance differences among CognoDB, Neo4j Aura, Memgraph, and FalkorDB under the selected graph workloads. FalkorDB recorded the lowest average latency for aggregation, indexed lookup, and 1-hop, 2-hop, and 3-hop traversal. It also achieved the highest measured throughput for the mixed read/write workload at all tested concurrency levels.

Neo4j Aura showed a consistently strong latency profile, while Memgraph and CognoDB recorded higher latency in the evaluated workloads. These findings represent the measured behavior under the stated experimental configuration and should be interpreted within those methodological limitations.

13. Project Artifacts

- Benchmark Python scripts
- Database connector implementations
- CSV benchmark result files
- Four-database comparison data
- Speedup calculations
- Performance visualization PNG files
- Final benchmark summary CSV files
- README project documentation

GitHub Repository: github.com/Hanifa246/four-database-graph-benchmark